

Appendix

Associations of long-term exposure to ambient PM_{2.5} with mortality in Chinese adults: A pooled analysis of cohorts in the China-PAR project

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Figure S1. Survey sites of the China-PAR project in 15 provinces

Blue stars represent survey sites in corresponding provinces

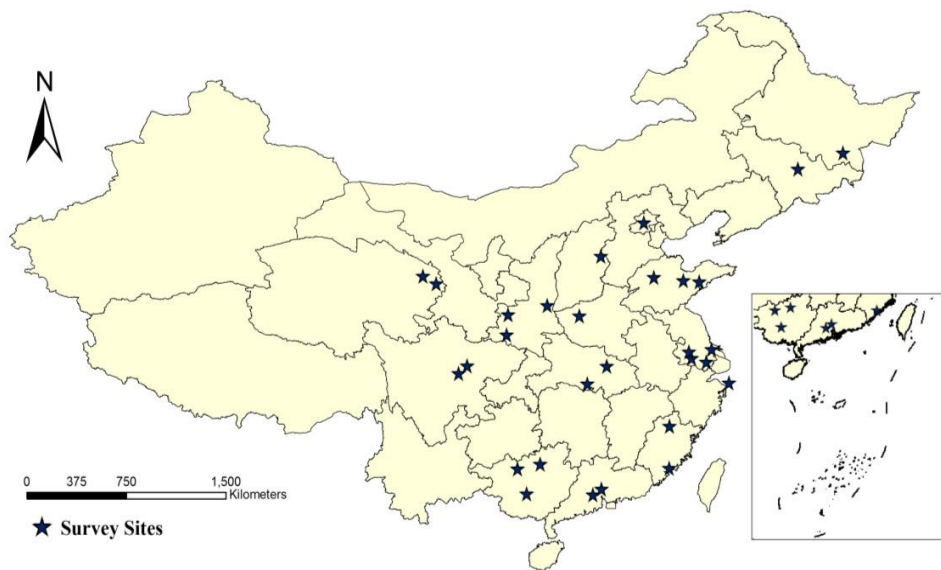


Figure S2. Flow chart of participants included and excluded in the study

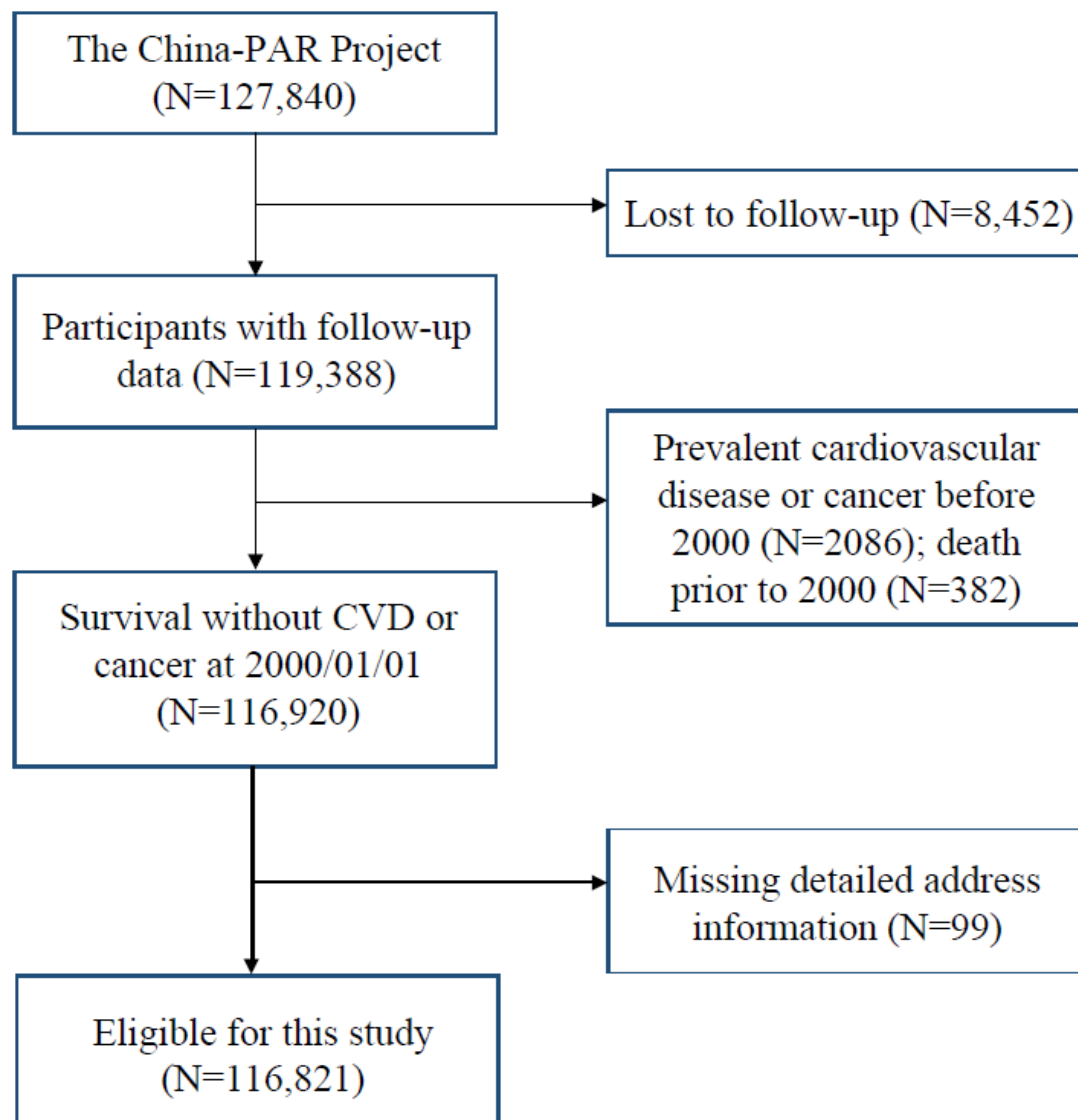


Figure S3. The overall and region-specific trends of annual PM_{2.5} levels in the study regions during 2000-2015

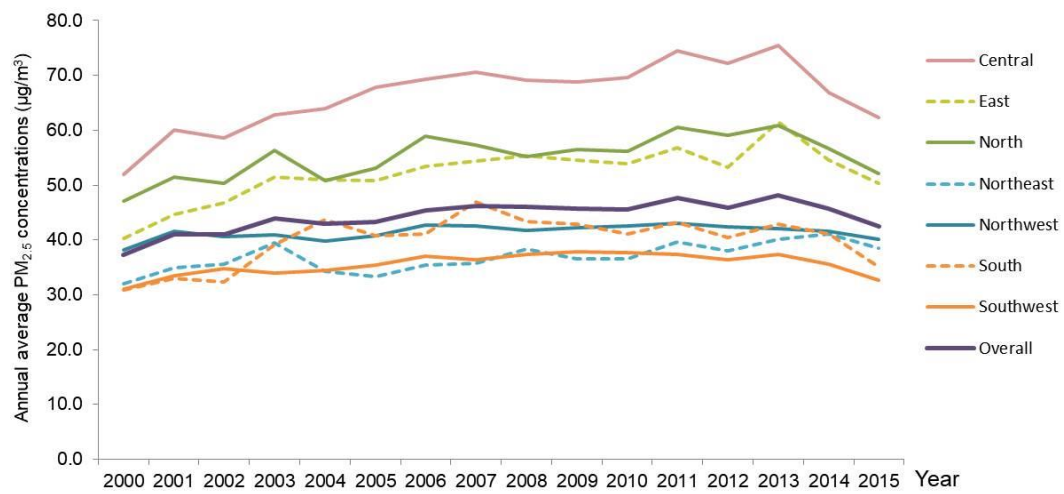


Table S1. Detailed cohort-specific follow-up information

	China MUCA (1992-1994)	China MUCA (1998)	InterASIA	CIMIC
Duration of follow-up, py ^a	151,330	128,769	152,992	469,533
NO. of lost to follow-up	267	985	1701	5499
Rate for lost to follow-up	1.9%	8.6%	10.9%	6.4%
NO. of events				
Non-accidental deaths	1334	590	1040	3431
cardio-metabolic deaths	396	167	467	1477

^a py, person-years

Table S2. Sensitivity analysis after excluding deaths within the first year for risk of death per 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$

	Hazard Ratio (95% CIs)	
	Non-accidental Deaths	Cardio-metabolic Deaths
Cases, n	6,040	2,367
Adjusted Model 1 ^a	1.11 (1.08-1.14)	1.28 (1.22-1.34)
Adjusted Model 2 ^b	1.13 (1.10-1.16)	1.32 (1.26-1.38)
Adjusted Model 3 ^c	1.13 (1.10-1.17)	1.26 (1.20-1.32)
Adjusted Model 4 ^d	1.12 (1.09-1.15)	1.23 (1.17-1.29)

^a Model 1: adjusted for age and sex.

^b Model 2: Model 1 + adjusted for education and urbanicity (urban vs. rural).

^c Model 3: Model 2 + adjusted for BMI, hypertension, diabetes, and total cholesterol.

^d Model 4: Model 3 + adjusted for smoking, drinking, and working-related physical activity at baseline.

Table S3. Sensitivity analysis using time-varying exposure at one-year time scale for risk of death per 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$

	Hazard Ratio (95% CIs)	
	Non-accidental Deaths	Cardio-metabolic Deaths
Adjusted Model 4 ^a	1.12 (1.09-1.15)	1.25 (1.19-1.30)

^a Adjusted for age, sex, education, urbanicity (urban vs. rural), BMI, hypertension, diabetes, total cholesterol, smoking, drinking, and working-related physical activity.

Table S4. Bayesian information criterion (BIC) values for different models

Models	BIC
Non-accidental death	
Cox model with ps ($PM_{2.5}$, $df=2$)	98,322.65
Cox model with ps ($PM_{2.5}$, $df=3$)	98,324.10
Cox model with ps ($PM_{2.5}$, $df=4$)	98,321.17
Cardio-metabolic death	
Cox model with ps ($PM_{2.5}$, $df=2$)	38,385.95
Cox model with ps ($PM_{2.5}$, $df=3$)	38,388.03
Cox model with ps ($PM_{2.5}$, $df=4$)	38,389.22

Table S5. Subgroup analysis for cardio-metabolic deaths associated with a 10 µg/m³ increase in PM_{2.5}

	Death	Person-year	HR (95% CI) ^a	<i>P</i> difference
Sex				
Male	1382	378,662.5	1.25 (1.17-1.33)	0.18
Female	1125	523,962.9	1.17 (1.09-1.25)	
Age, yrs				
<60	845	706,923.8	1.16 (1.07-1.26)	0.75
≥60	1662	195,638.7	1.18 (1.11-1.24)	
Urbanicity				
Rural	2216	736,094.3	1.24 (1.18-1.30)	0.91
Urban	291	166,531.0	1.21 (0.84-1.74)	
BMI, kg/m ²				
<25	1757	622,080.9	1.20 (1.13-1.27)	0.29
≥25	745	280,063.0	1.27 (1.16-1.38)	
Smoking				
Nonsmoker	1585	632,577.0	1.21 (1.15-1.28)	0.93
Smoker	914	266,373.5	1.21 (1.12-1.31)	
Drinking				
Excess	189	71,241.5	1.37 (1.16-1.62)	0.240
Moderate	125	63,863.6	1.36 (1.10-1.70)	0.373
None	2012	712,641.3	1.19 (1.13-1.25)	
Hypertension				
Stage2 HTN	1012	142,928.4	1.27 (1.19-1.36)	0.044
Stage1 HTN	484	119,114.8	1.19 (1.07-1.32)	0.264
Prehypertension	619	297,716.4	1.27 (1.15-1.40)	0.078
Normal	391	342,561.6	1.06 (0.93-1.22)	
Diabetes				
Yes	363	41,971.6	1.16 (1.03-1.32)	0.47

No	2000	805,092.3	1.22 (1.17-1.29)
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Abbreviations: BMI, body mass index; HTN, hypertension; HR, hazard ratio.

^a The HRs (95% CIs) were the results after adjustment of age, sex, education, urbanicity (urban vs. rural), BMI, hypertension, diabetes, total cholesterol, smoking, drinking, and working-related physical activity at baseline.

Table S6. Sensitivity analysis using the cohort as a covariate for risks of mortality per 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$

	Hazard Ratio (95% CIs)	
	Non-accidental Deaths	Cardio-metabolic Deaths
Multivariate adjustment ^a	1.08 (1.04-1.10)	1.18 (1.13-1.23)

^a Adjusted for covariates in Model 4 (i.e., age, sex, education, urbanicity (urban vs. rural), BMI, hypertension, diabetes, total cholesterol, smoking, drinking, and working-related physical activity) and cohort.

Table S7. Hazard ratio of mortality associated with a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ using the exposure data before 2013

	Hazard Ratio (95% CIs)	
	Non-accidental Deaths	Cardio-metabolic Deaths
Multivariable adjustment ^a	1.11 (1.08, 1.15)	1.23 (1.17, 1.29)

^a Adjusted for covariates in Model 4 (i.e., age, sex, education, urbanicity (urban vs. rural), BMI, hypertension, diabetes, total cholesterol, smoking, drinking, and working-related physical activity).

Table S8. Hazard ratio of mortality associated with a 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ after adjustment for neighborhood education years

	Hazard Ratio (95% CIs)	
	Non-accidental Deaths	Cardio-metabolic Deaths
Multivariable adjustment ^a	1.13 (1.09, 1.16)	1.24 (1.19, 1.30)

^a Adjusted for covariates in Model 4 (i.e., age, sex, education, urbanicity (urban vs. rural), BMI, hypertension, diabetes, total cholesterol, smoking, drinking, and working-related physical activity) and neighborhood education years.

Table S9. Hazard ratios (95% CI) according to quartiles of PM_{2.5} levels

	PM _{2.5} categories [*]				
	Q1	Q2	Q3	Q4	<i>P</i> _{trend}
Non-accidental mortality					
deaths, n	2086	1312	1581	1416	
age-, sex-adjusted rate [§]	643	734	902	556	
Hazard Ratio (95% CI)					
Model 1 ^a	1.00	0.98(0.90,1.07)	1.16(1.05,1.28)	1.43(1.30,1.57)	<i>P</i> <0.001
Model 2 ^b	1.00	0.98(0.89,1.07)	1.2(1.09,1.32)	1.49(1.35,1.64)	<i>P</i> <0.001
Model 3 ^c	1.00	0.96(0.87,1.06)	1.18(1.06,1.31)	1.45(1.31,1.61)	<i>P</i> <0.001
Model 4 ^d	1.00	0.92(0.84,1.02)	1.09(0.98,1.22)	1.36(1.23,1.51)	<i>P</i> <0.001
Cardiometabolic mortality					
deaths, n	634	498	703	672	
age-, sex-adjusted rate [§]	193	294	398	240	
Hazard Ratio (95% CI)					
Model 1 ^a	1.00	1.05(0.90,1.23)	1.66(1.41,1.96)	2.19(1.87,2.56)	<i>P</i> <0.001
Model 2 ^b	1.00	1.06(0.91,1.24)	1.75(1.48,2.06)	2.32(1.98,2.71)	<i>P</i> <0.001
Model 3 ^c	1.00	0.99(0.84,1.16)	1.48(1.24,1.76)	1.92(1.63,2.28)	<i>P</i> <0.001
Model 4 ^d	1.00	0.94(0.79,1.11)	1.36(1.13,1.62)	1.75(1.48,2.08)	<i>P</i> <0.001

*PM_{2.5} categories: minimum, 31.2 µg/m³; 25th percentile, 54.5µg/m³; 50th percentile, 59.5µg/m³; 75th percentile, 78.2µg/m³; maximum, 97.0µg/m³.

[§]Age- and sex-adjusted mortality rate per 100,000 person-years

^a Model 1: adjusted for age and sex.

^b Model 2: Model 1 + adjusted for education and urbanicity (urban vs. rural).

^c Model 3: Model 2 + adjusted for BMI, hypertension, diabetes, and total cholesterol.

^d Model 4: Model 3 + adjusted for smoking, drinking, and working-related physical activity at baseline.

Table S10. Comparison of baseline characteristics among participants included with those among participants lost to follow-up ^a

	Mean $\pm$ SD or %	
	Included	Loss to follow-up
Participants (n)	116,821	8,452
Age (y)	51.6 $\pm$ 11.7	47.4 $\pm$ 13.8
Women, %	59.0	57.3
Urban, %	12.3	23.9
Education ($\geq$ high school level), %	14.2	28.8
Working PA ($\geq$ medium level), %	55.7	63.2
Smoking, %	27.5	27.4
Alcohol drinkers, %	19.6	16.9
BMI (kg/m ²)	23.6 $\pm$ 3.6	23.6 $\pm$ 3.6
SBP (mmHg)	127.5 $\pm$ 21.1	122.7 $\pm$ 19.6
DBP (mmHg)	78.8 $\pm$ 11.7	76.8 $\pm$ 11.4
Total cholesterol (mg/dL)	175.2 $\pm$ 35.5	170.8 $\pm$ 35.7
Hypertension, %	31.8	23.9
Diabetes, %	5.5	6.2

Abbreviations: SD, standard derivation; Working PA, working-related physical activity; BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure.

^a Baseline characteristics are presented as mean $\pm$ standard deviation for continuous variables or percentage for categorical variables.